

EduPro Instructor Performance and Course Quality Evaluation

Submitted by:

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Internship Project Report
(Research Paper)

Organization:

Unified Mentor

Domain:

Data Analytics

Tools & Technologies Used:

Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab,
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ABSTRACT

The quality of online education depends significantly on instructor effectiveness, course structure, and learner engagement. Educational platforms require data-driven approaches to evaluate teaching performance and improve course quality. This study analyzes instructor performance and course quality within the EduPro educational platform using exploratory data analysis and interactive dashboard visualization techniques.

The project examines instructor ratings, years of experience, expertise domains, course categories, course levels, enrollment behavior, and learner interactions. Multiple analytical techniques including descriptive statistics, correlation analysis, comparative analysis, and visualization methods were applied to identify patterns and trends.

The findings reveal that teaching experience positively influences instructor ratings, while expertise domains demonstrate varying levels of learner satisfaction and enrollment performance. The study also highlights differences in course quality across categories and learning levels. An interactive Streamlit dashboard was developed to support real-time exploration of key performance indicators and educational insights.

The results provide actionable recommendations for improving instructor evaluation, course development, learner satisfaction, and strategic decision-making within online learning platforms.

Keywords: Online Education, Instructor Performance, Course Quality, Data Analytics, Educational Dashboard, Learner Engagement, Streamlit.

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INTRODUCTION

Online learning platforms have transformed the educational landscape by providing learners with flexible access to high-quality educational content. As the number of online courses continues to grow, ensuring instructional quality and learner satisfaction has become increasingly important.

Instructor performance plays a critical role in determining educational effectiveness. Highly rated instructors often contribute to better learning outcomes, improved engagement, and stronger course completion rates. Similarly, course quality influences learner satisfaction, enrollment decisions, and platform reputation.

The EduPro platform offers courses across multiple disciplines including Data Science, Artificial Intelligence, Machine Learning, Finance, Marketing, Cybersecurity, and Web Development. With a large number of instructors and learners, it becomes essential to establish a systematic framework for evaluating teaching effectiveness and course quality.

This project utilizes data analytics techniques to analyze instructor performance, identify key factors influencing ratings and enrollments, and provide actionable recommendations for educational improvement. The study also includes the development of an interactive Streamlit dashboard for visualizing performance metrics and supporting data-driven decision-making.

PROBLEM STATEMENT

Educational institutions and online learning platforms often face challenges in evaluating instructor performance and course quality objectively. Traditional evaluation approaches may not fully capture patterns related to learner engagement, instructor expertise, course effectiveness, and enrollment behavior.

Without comprehensive analytical insights, organizations may struggle to:

- Identify high-performing instructors.
- Detect areas requiring instructional improvement.
- Understand enrollment trends and learner preferences.
- Improve course quality across categories and levels.
- Support strategic educational planning.

Therefore, there is a need for a data-driven evaluation framework capable of analysing instructor performance, course quality, and learner engagement to facilitate informed decision-making and continuous improvement.

RESEARCH OBJECTIVES

The primary objective of this study is to evaluate instructor performance and course quality on the EduPro platform using data analytics techniques. The specific objectives are as follows:

1. To analyze instructor ratings and assess overall teaching performance across the platform.
2. To examine the relationship between teaching experience and instructor ratings.
3. To identify expertise domains associated with higher instructional effectiveness and learner satisfaction.
4. To evaluate course quality across different course categories and learning levels.
5. To analyze learner enrollment patterns and identify factors influencing course participation.
6. To investigate the relationship between instructor ratings and course ratings.
7. To compare instructor performance across demographic and professional characteristics.
8. To identify top-performing and low-performing instructors based on rating metrics.
9. To develop an interactive Streamlit dashboard for visualizing key performance indicators and educational insights.
10. To provide data-driven recommendations for improving teaching quality, learner engagement, and overall platform performance.

DATASET DESCRIPTION

The EduPro project utilizes four interconnected datasets that collectively represent instructors, courses, learners, and enrollment transactions. These datasets were analyzed to evaluate instructor effectiveness, course quality, and learner engagement patterns.

1. Teachers Dataset -

The Teachers dataset contains demographic and professional information about instructors on the EduPro platform.

Attributes -

- Teacher ID
- Teacher Name
- Age
- Gender
- Expertise
- Years of Experience
- Teacher Rating

This dataset was primarily used to analyze instructor performance, expertise-based comparisons, rating distributions, and experience-related trends.

2. Courses Dataset -

The Courses dataset contains information related to educational offerings available on the platform.

Attributes -

- Course ID
- Course Name

- Course Category
- Course Level
- Course Price
- Course Duration
- Course Rating

This dataset was used to evaluate course quality, learner satisfaction, pricing strategies, and category-level performance.

3. Transactions Dataset -

The Transactions dataset stores learner enrollment and payment records.

Attributes -

- Transaction ID
- User ID
- Teacher ID
- Course ID
- Transaction Date
- Amount
- Payment Method

This dataset was used to analyze enrollment behavior and identify trends related to instructor popularity and learner engagement.

4. Users Dataset -

The Users dataset contains information about learners enrolled on the platform.

Attributes -

- User ID
- User Name
- Age

- Gender
- Email

This dataset supports learner demographic analysis and helps understand the user base of the platform.

Dataset Summary -

Dataset	Purpose
Teachers	Instructor Performance Analysis
Courses	Course Quality Evaluation
Transactions	Enrollment Analysis
Users	Learner Demographics Analysis

The combination of these datasets enables a comprehensive evaluation of educational quality, instructor effectiveness, and learner engagement.

METHODOLOGY

1. Data Collection

The dataset used in this project was obtained from the EduPro educational platform. The data consists of four interconnected datasets namely Teachers, Courses, Transactions, and Users. These datasets provide information regarding instructor profiles, course characteristics, learner details, and enrollment activities.

The collected data was utilized to evaluate instructor performance, course quality, learner engagement, and enrollment trends across the platform.

2. Data Preparation

The datasets were imported into Google Colab using the Pandas library. Before performing analysis, the data was examined for missing values, duplicate records, and data inconsistencies.

Data preparation activities included:

- Loading Excel sheets into separate data frames.
- Verifying data types and column structures.
- Checking for missing and duplicate values.
- Merging related datasets using Teacher ID and Course ID.
- Creating integrated datasets for advanced analysis.

These preprocessing steps ensured data quality and analytical reliability.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of instructors, courses, and enrollment behavior.

The analysis focused on:

- Teacher rating distribution.

- Age distribution of instructors.
- Experience distribution analysis.
- Gender-based rating comparisons.
- Expertise-wise performance evaluation.
- Course category performance analysis.
- Course level comparison.
- Enrollment trend analysis.
- Relationship between teacher ratings and course ratings.

EDA enabled the identification of meaningful trends and patterns within the educational ecosystem.

4. Data Integration

To evaluate instructor impact on course quality and enrollment performance, multiple datasets were integrated.

The Transactions dataset was merged with the Teachers dataset using Teacher ID. Similarly, course-related information was connected using Course ID.

This integration enabled:

- Enrollment analysis by instructor.
- Expertise-based enrollment analysis.
- Teacher rating versus enrollment analysis.
- Teacher rating versus course rating analysis.

The integrated dataset provided a holistic view of educational performance.

5. Statistical Analysis

Several statistical measures were applied to summarize and evaluate the dataset.

The analysis included:

- Mean

- Median
- Standard Deviation
- Correlation Analysis
- Frequency Distribution

Correlation analysis was specifically used to examine relationships between instructor ratings, course ratings, and experience levels.

These measures supported objective interpretation of educational performance indicators.

6. Data Visualization

Data visualization techniques were employed to communicate findings effectively.

The visualizations used in the project include:

Histograms -

Used to analyze:

- Teacher Rating Distribution
- Teacher Age Distribution

Scatter Plots -

Used to analyze:

- Experience vs Teacher Rating
- Teacher Rating vs Enrollment
- Course Price vs Rating

Bar Charts -

Used to analyze:

- Expertise-wise Average Rating
- Course Category Rating
- Course Level Rating

- Enrollment by Expertise

Boxplots -

Used to compare:

- Teacher Rating by Gender

Heatmaps -

Used to visualize:

- Course Quality across Categories and Levels
- Correlation among Teacher Metrics

These visualizations improved analytical understanding and supported decision-making.

7. Dashboard Development -

An interactive Streamlit dashboard was developed to provide real-time educational insights.

The dashboard includes:

- Key Performance Indicators (KPIs)
- Expertise Filters
- Gender Filters
- Course Category Filters
- Course Level Filters
- Rating Range Selection
- Interactive Visualizations
- Dataset Preview
- Dataset Download Options

The dashboard allows users to dynamically explore instructor performance, course quality, and enrollment behavior.

8. Tools and Technologies Used –

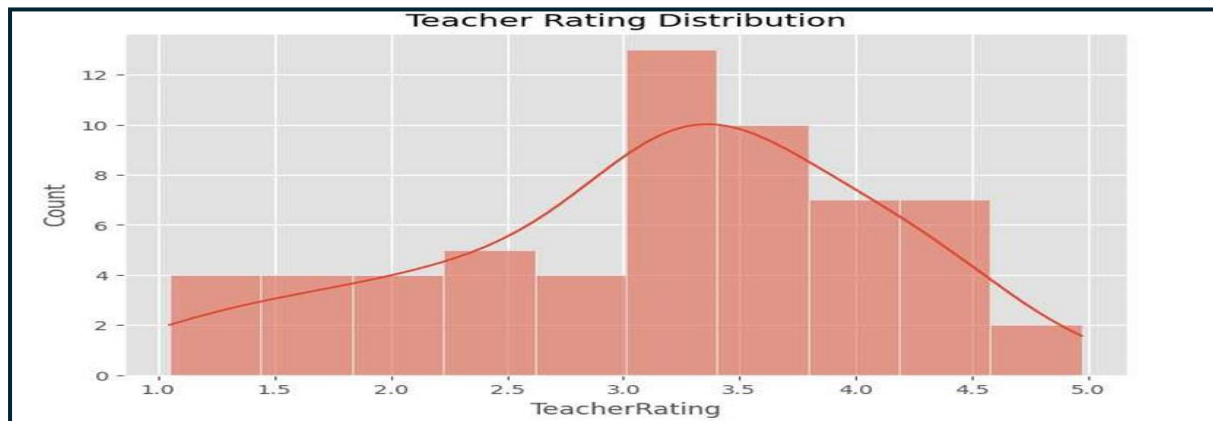
The following technologies were utilized throughout the project:

Tool	Purpose
Python	Data Analysis
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Google Colab	Development Environment
Streamlit	Dashboard Development
GitHub	Version Control & Deployment
Microsoft Word	Research Paper Documentation

These technologies collectively supported data processing, visualization, dashboard creation, and project documentation.

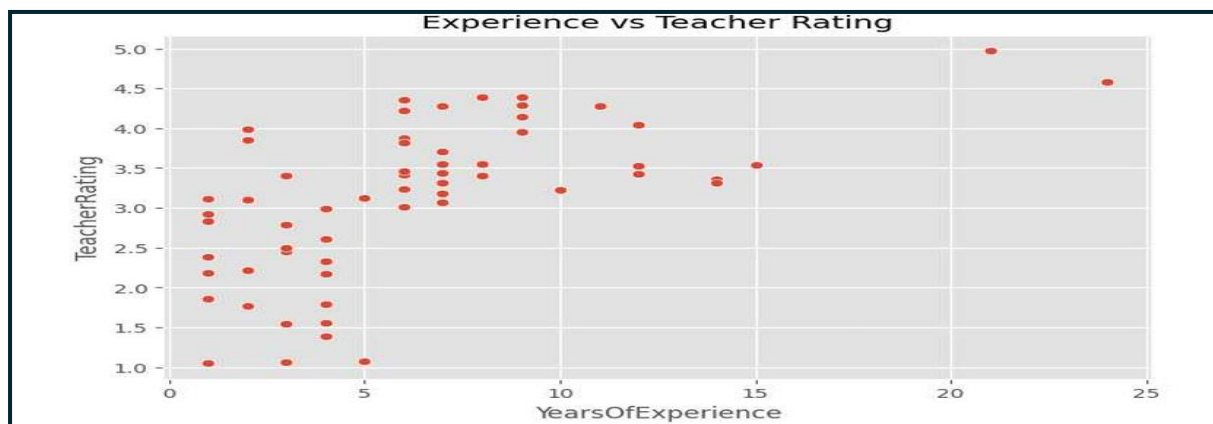
EXPLORATORY DATA ANALYSIS (EDA)

Figure 1: Teacher Rating Distribution



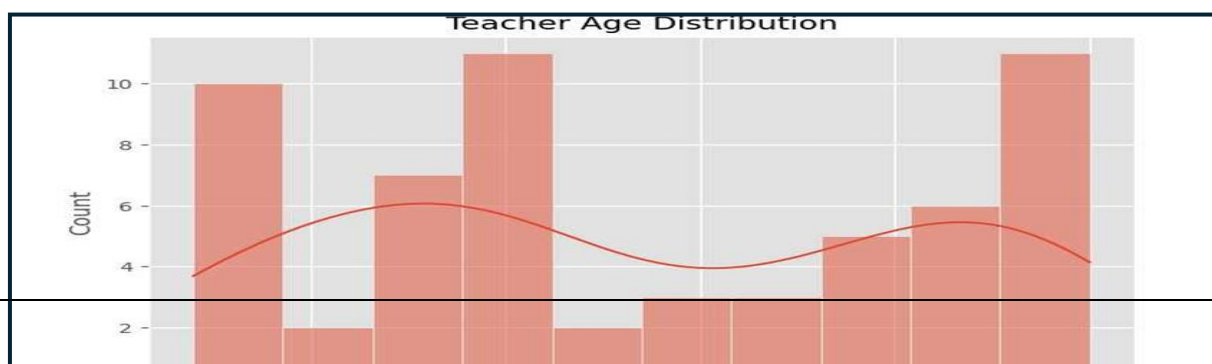
Insight: The majority of instructors received ratings between 3.0 and 4.0, indicating generally satisfactory teaching performance across the EduPro platform. Only a limited number of instructors achieved exceptionally high or low ratings.

Figure 2: Experience vs Teacher Rating



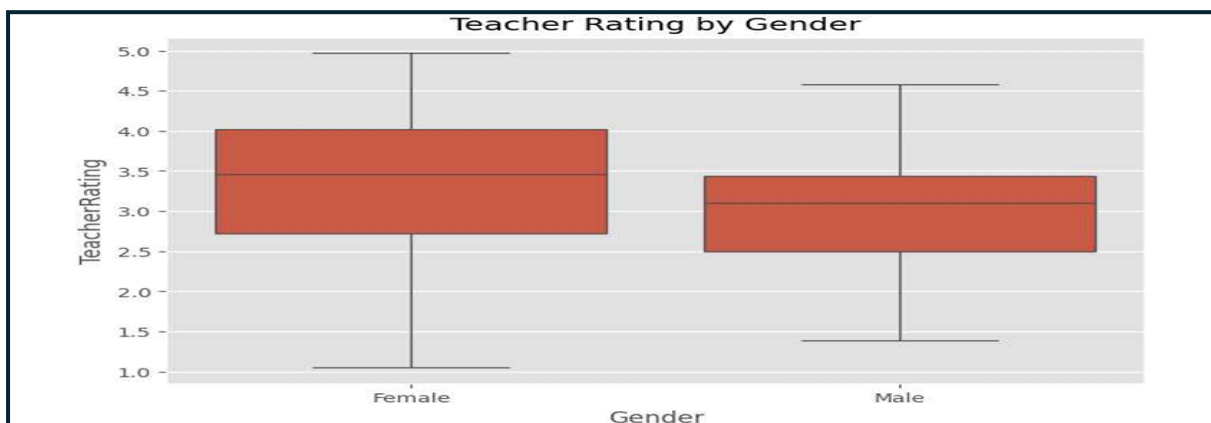
Insight: A positive relationship is observed between teaching experience and instructor ratings. Instructors with greater experience tend to receive higher ratings, suggesting that experience contributes positively to teaching effectiveness.

Figure 3: Teacher Age Distribution



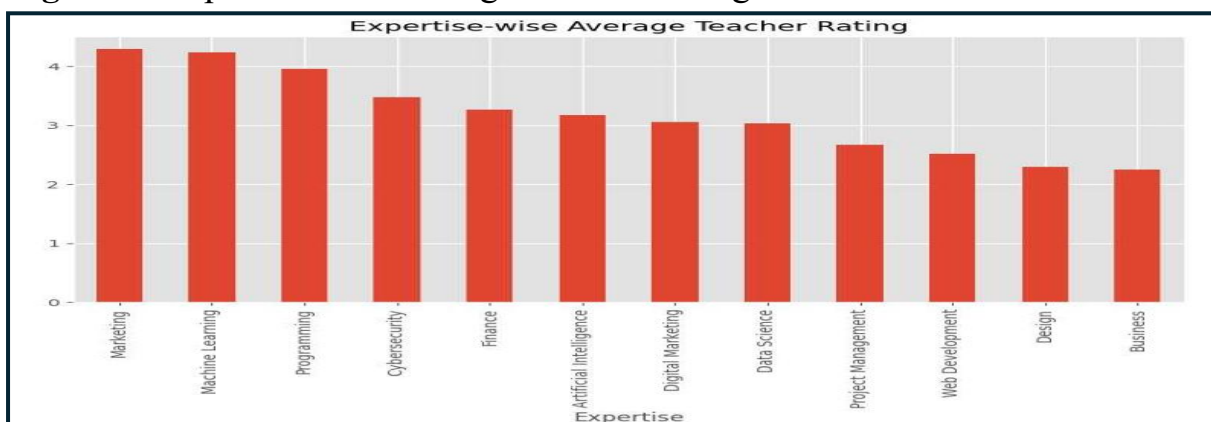
Insight: The age distribution shows a balanced workforce consisting of both young and experienced educators. Most instructors fall within the age range of 30 to 50 years, indicating a diverse teaching population.

Figure 4: Teacher Rating by Gender



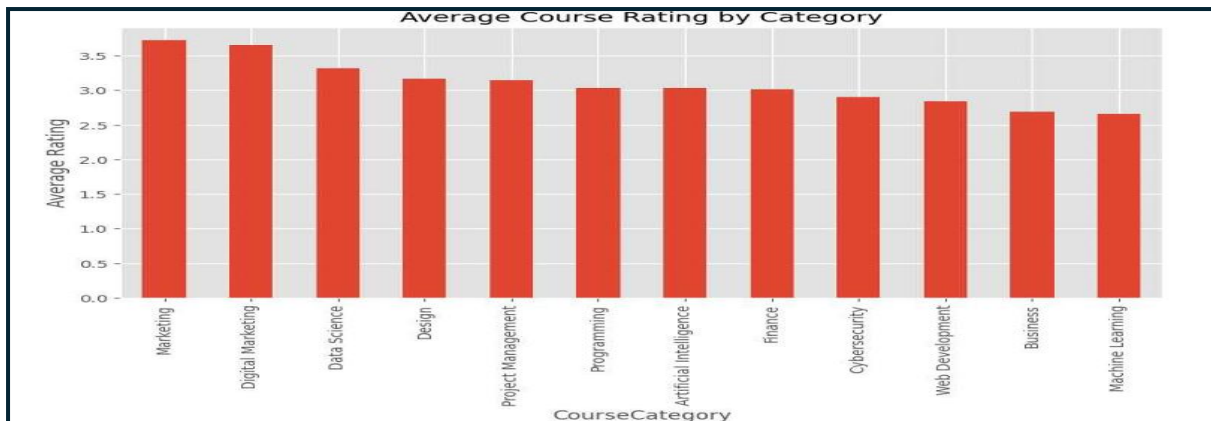
Insight: The analysis shows relatively similar rating distributions across genders. Female instructors demonstrate a slightly higher median rating, although overall performance remains comparable between both groups.

Figure 5: Expertise-wise Average Teacher Rating



Insight: Marketing, Machine Learning, and Programming instructors exhibit the highest average ratings. This suggests stronger learner satisfaction and instructional effectiveness within these expertise domains.

Figure 6: Average Course Rating by Category



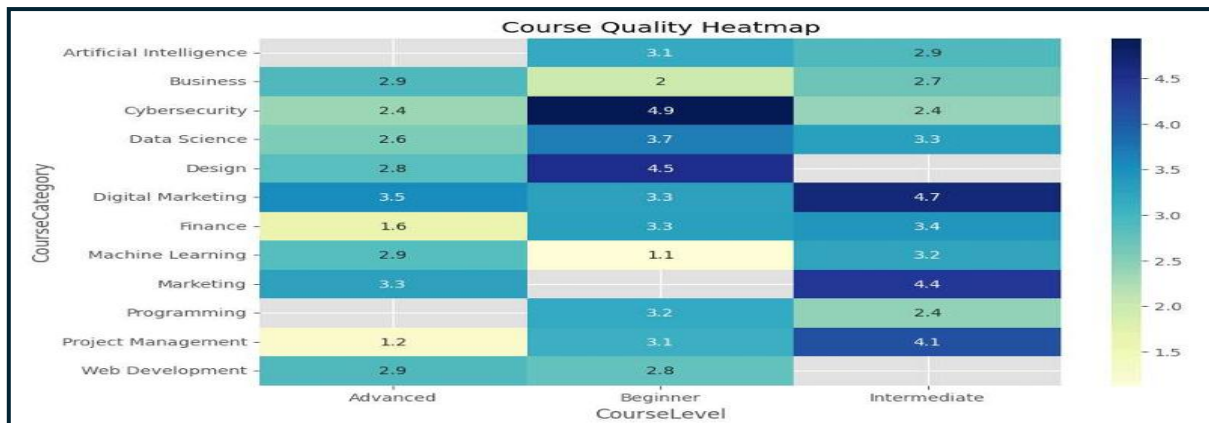
Insight: Marketing and Digital Marketing courses achieved the highest average ratings, indicating strong learner engagement and course quality. Business and Machine Learning courses received comparatively lower ratings.

Figure 7: Average Course Rating by Level



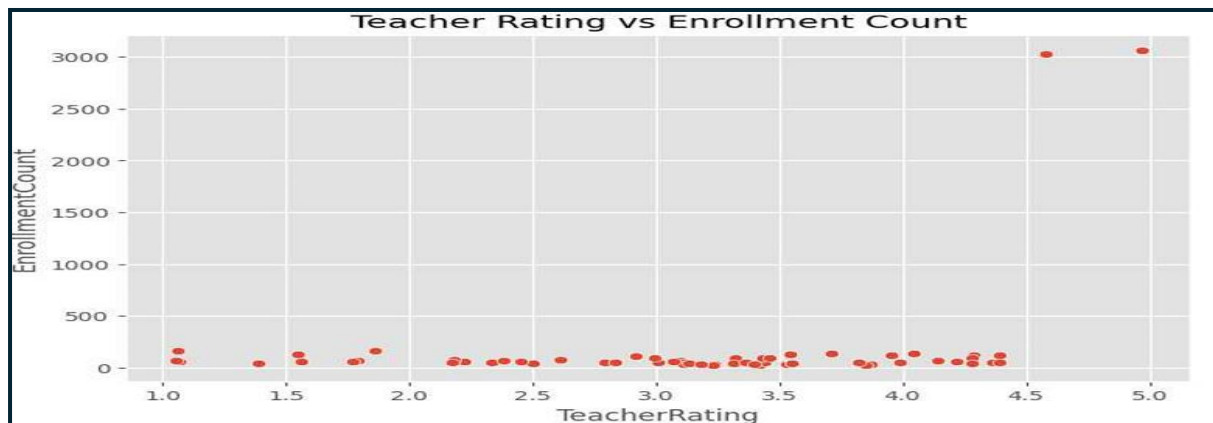
Insight: Intermediate-level courses received the highest average ratings among all course levels. This suggests that learners find intermediate courses more engaging and aligned with their learning requirements.

Figure 8: Course Quality Heatmap



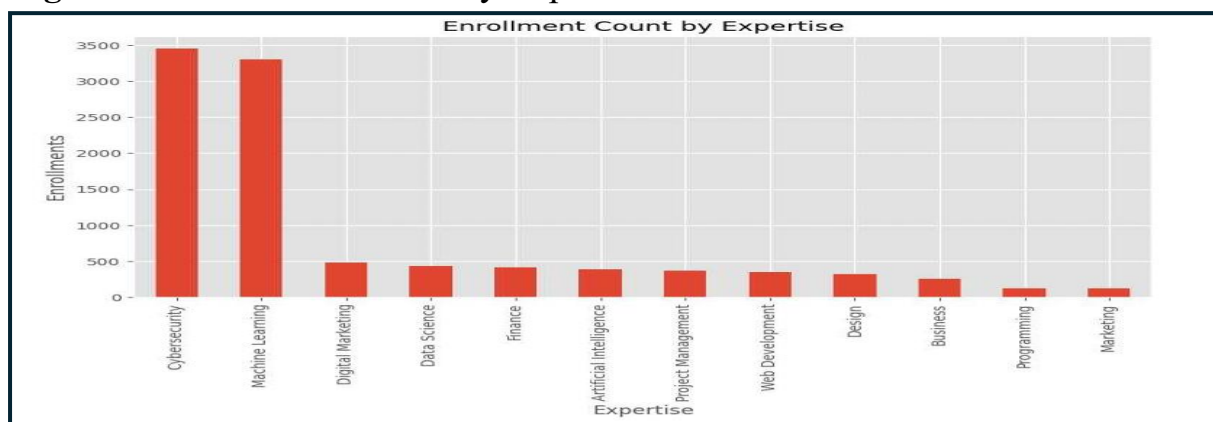
levels, while others demonstrate opportunities for quality improvement.

Figure 9: Teacher Rating vs Enrollment Count



Insight: A positive relationship exists between teacher ratings and enrollment counts. Higher-rated instructors generally attract more learners, indicating that instructor reputation influences enrollment decisions.

Figure 10: Enrollment Count by Expertise



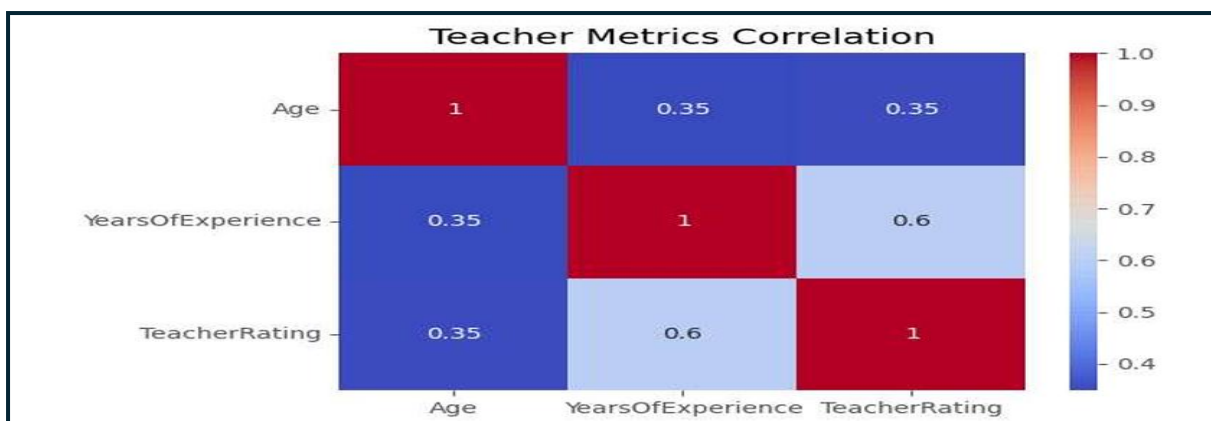
Insight: Cybersecurity and Machine Learning generated the highest enrollment counts among all expertise areas. This reflects strong market demand for technical and emerging technology skills.

Figure 11: Course Price vs Course Rating



Insight: The scatter plot indicates no significant relationship between course pricing and learner ratings. Higher-priced courses do not necessarily receive better evaluations, suggesting that quality perception depends on multiple factors beyond cost.

Figure 12: Teacher Metrics Correlation Analysis



Insight: The correlation analysis reveals a moderate positive relationship (0.60) between years of experience and teacher ratings. This suggests that experienced instructors generally achieve better performance outcomes and higher learner satisfaction.

RESULTS & KEY FINDINGS

The analysis of the EduPro dataset generated several important insights regarding instructor performance, course quality, and learner engagement.

Key Finding 1: Overall Instructor Performance -

The average teacher rating across the platform was approximately 3.12, indicating moderate overall teaching performance. Most instructors received ratings within the satisfactory range, demonstrating consistent instructional quality across the platform.

Key Finding 2: Experience Influences Teacher Ratings -

The analysis revealed a positive relationship between years of experience and teacher ratings. Experienced instructors generally achieved higher ratings, suggesting that professional experience contributes positively to teaching effectiveness and learner satisfaction.

Key Finding 3: Expertise Impacts Performance -

Significant differences were observed across expertise domains. Marketing, Machine Learning, and Programming instructors achieved higher average ratings compared to other expertise areas, indicating stronger learner satisfaction within these domains.

Key Finding 4: Course Quality Varies Across Categories -

Course ratings differed across educational categories and learning levels. Marketing and Digital Marketing courses received the highest average ratings, while certain categories demonstrated opportunities for further quality enhancement.

Key Finding 5: Intermediate Courses Perform Best -

Among all course levels, Intermediate courses achieved the highest average ratings. This suggests that learners perceive intermediate-level content as highly relevant and engaging.

Key Finding 6: Teacher Ratings Influence Enrollment -

The analysis showed that higher-rated instructors generally attracted greater enrollment volumes. This indicates that instructor reputation plays an important role in learner enrollment decisions.

Key Finding 7: High Demand for Technical Domains -

Cybersecurity and Machine Learning generated the highest enrollment counts among all expertise domains. This reflects strong industry demand for technology-focused skills and certifications.

Key Finding 8: Course Pricing Has Limited Impact on Ratings -

No significant relationship was identified between course prices and learner ratings. Higher-priced courses did not necessarily receive better evaluations, suggesting that perceived quality is influenced by content and instructional effectiveness rather than pricing.

Key Finding 9: Teacher Rating vs Course Rating Relationship -

The correlation between Teacher Rating and Course Rating was approximately -0.002, indicating virtually no relationship between the two variables. This finding suggests that course ratings are influenced by multiple factors beyond instructor performance, including course design, content quality, learner engagement, and curriculum structure.

Key Finding 10: Teacher Metrics Correlation Analysis -

The correlation analysis revealed a moderate positive relationship (0.60) between years of experience and teacher ratings. This further confirms the importance of experience in achieving better instructional outcomes.

Summary -

The findings indicate that instructor experience, expertise domain, and reputation significantly influence educational outcomes and learner engagement. The analysis also highlights the importance of course quality, content design, and market demand in shaping learner satisfaction and enrollment behavior.

DASHBOARD OVERVIEW

An interactive analytics dashboard was developed using Streamlit to provide real-time insights into instructor performance, course quality, learner engagement, and enrollment trends within the EduPro platform.

The dashboard was designed to transform analytical findings into an easy-to-use decision-support system for educational stakeholders. Through interactive filters and visualizations, users can dynamically explore educational performance metrics and identify patterns across multiple dimensions.

Dashboard Features

The dashboard includes several analytical components that support comprehensive educational performance evaluation:

Key Performance Indicators (KPIs)

The KPI section provides a quick overview of important performance metrics, including:

- Average Teacher Rating
- Average Course Rating
- Average Teaching Experience
- Total Number of Teachers
- Total Number of Courses
- Total Number of Transactions

These metrics help users quickly understand the overall status of the platform.

Interactive Filters

The dashboard allows users to customize analysis through interactive filters, including:

- Expertise Filter

- Gender Filter
- Course Category Filter
- Course Level Filter
- Teacher Rating Range Filter

These filters enable focused exploration of specific instructor groups and course segments.

Analytical Visualizations

The dashboard contains multiple visualizations to support educational analysis:

- Teacher Rating Distribution
- Experience vs Teacher Rating Analysis
- Teacher Age Distribution
- Teacher Rating by Gender
- Expertise-wise Average Rating
- Course Category Analysis
- Course Level Analysis
- Course Quality Heatmap
- Teacher Rating vs Enrollment Analysis
- Enrollment by Expertise
- Course Price vs Rating Analysis

These visualizations help identify trends, relationships, and performance differences across the platform.

Instructor Performance Analysis

The dashboard highlights top-performing and low-performing instructors through dedicated ranking tables.

This feature assists educational administrators in identifying best-performing educators and areas requiring improvements.

Dataset Download Functionality

Users can download filtered datasets directly from the dashboard for further analysis and reporting purposes.

This functionality improves accessibility and supports data-driven decision-making.

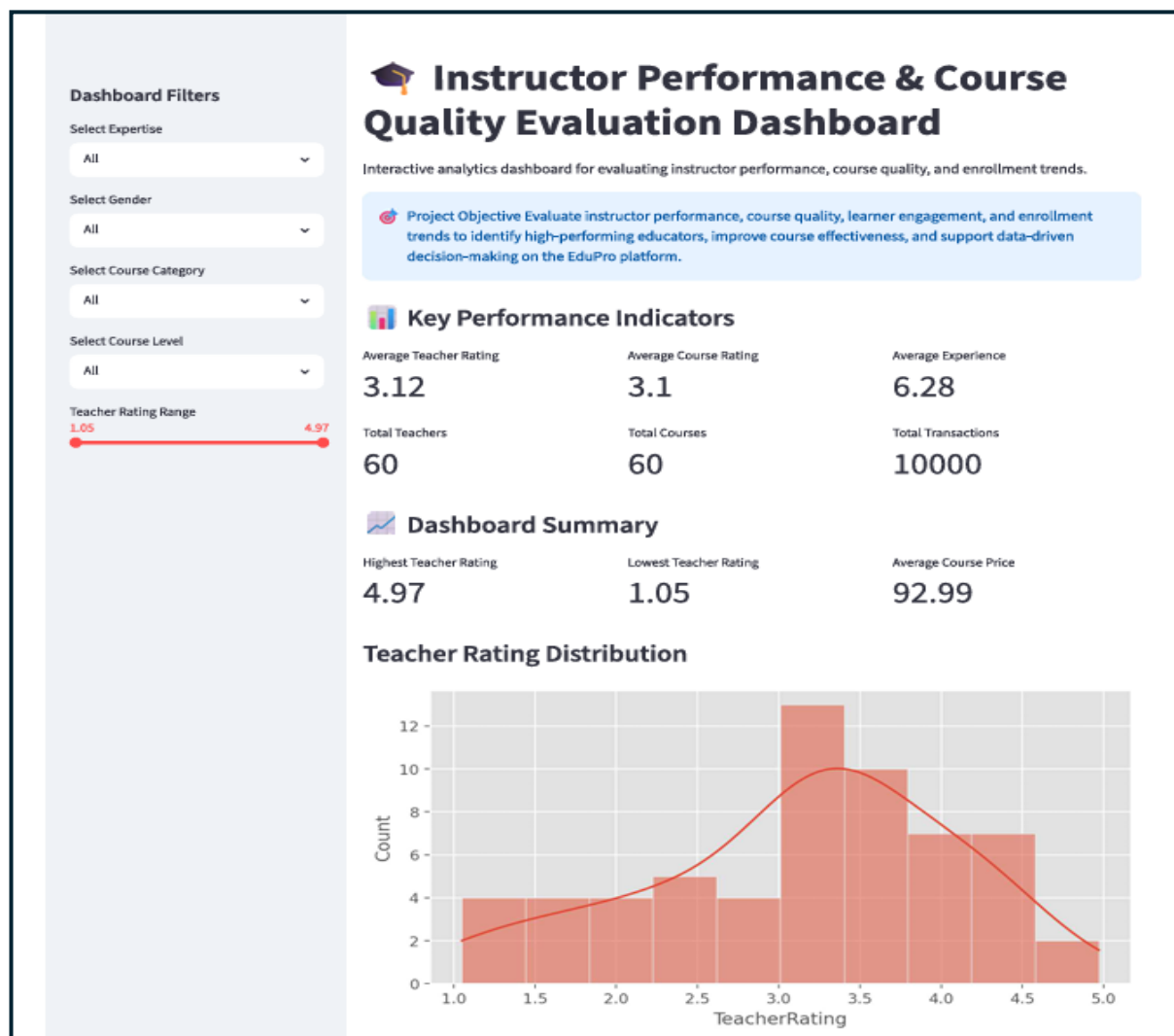
Dashboard Deployment

The dashboard was successfully deployed using Streamlit Cloud, enabling online access and real-time interaction without requiring local software installation.

The deployment ensures scalability, accessibility, and ease of use for educational stakeholders.

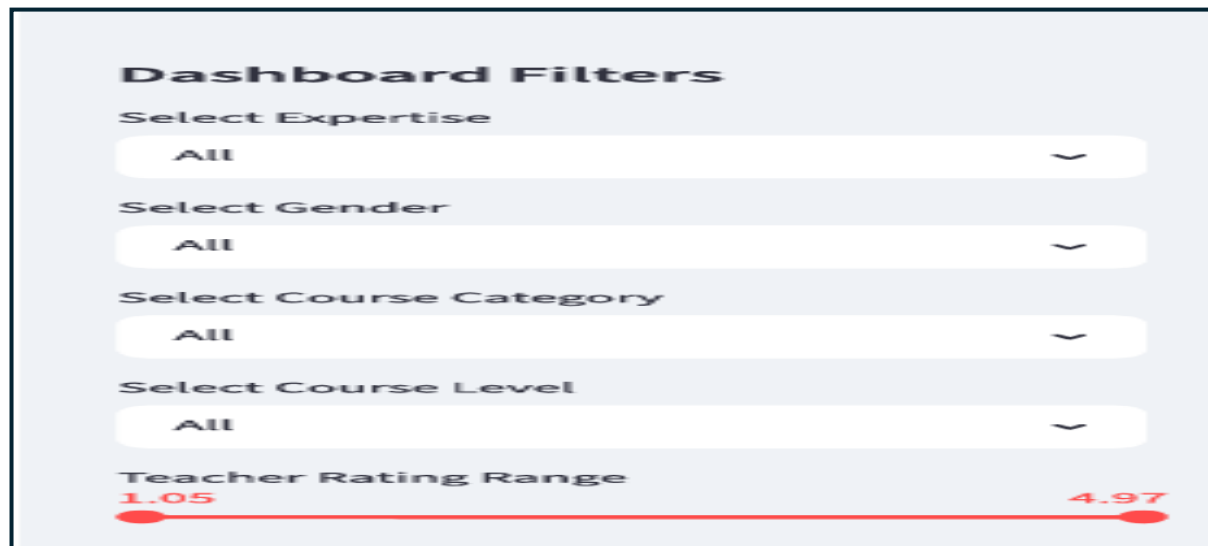
Dashboard Screenshots

Figure 13: Dashboard Home Page



Insight: The dashboard homepage provides a comprehensive overview of instructor performance and course quality metrics. It includes interactive filters, key performance indicators, and analytical visualizations that support data-driven decision-making.

Figure 14: Interactive Filter Panel

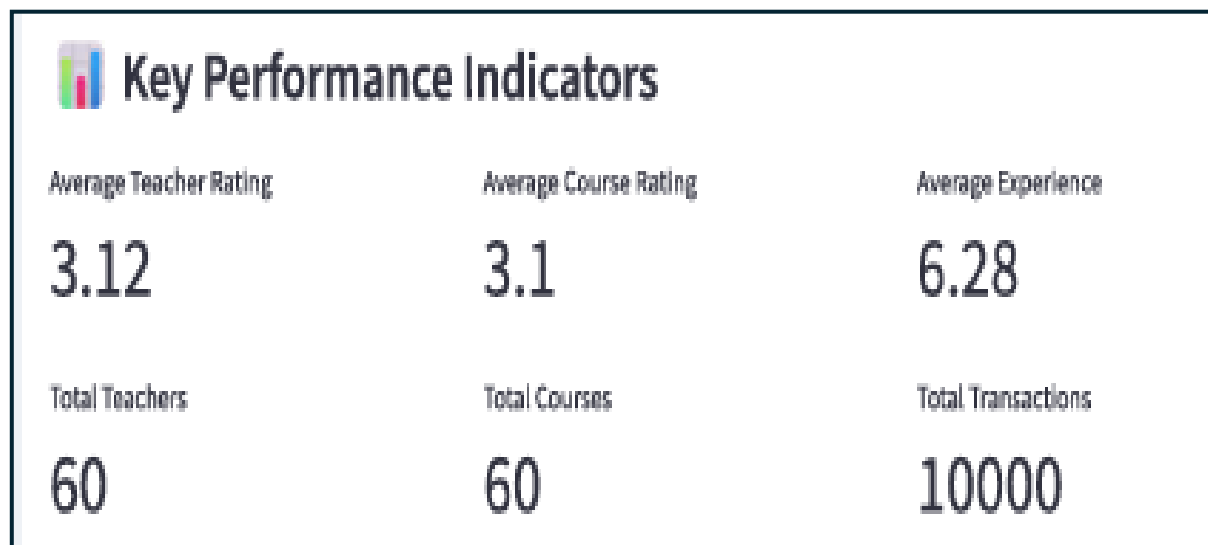


The figure shows a 'Dashboard Filters' panel with five filter sections. Each section has a label and a dropdown menu. The first four sections are 'Select Expertise', 'Select Gender', 'Select Course Category', and 'Select Course Level', all with 'All' selected. The fifth section is 'Teacher Rating Range', which is a range slider with a red line and dots at 1.05 and 4.97.

Filter Type	Value
Select Expertise	All
Select Gender	All
Select Course Category	All
Select Course Level	All
Teacher Rating Range	1.05 to 4.97

Insight: The interactive filter panel enables users to customize analysis based on expertise, gender, course category, course level, and teacher rating range. This functionality improves analytical flexibility and user experience.

Figure 15: Key Performance Indicators (KPI)



The figure shows a 'Key Performance Indicators' dashboard with a grid of six metrics. The first row contains 'Average Teacher Rating' (3.12), 'Average Course Rating' (3.1), and 'Average Experience' (6.28). The second row contains 'Total Teachers' (60), 'Total Courses' (60), and 'Total Transactions' (10000). Each metric is represented by a small bar chart icon and a large numerical value.

Key Performance Indicators		
Average Teacher Rating	Average Course Rating	Average Experience
3.12	3.1	6.28
Total Teachers	Total Courses	Total Transactions
60	60	10000

Insight: The KPI section summarizes important platform metrics, including average teacher rating, average course rating, average teaching experience, total teachers, total courses, and total transactions.

Figure 16: Analytical Visualizations



Insight: The dashboard provides multiple visualizations that help users identify trends, relationships, and performance patterns across instructors and courses. These visualizations support comprehensive educational performance analysis.

RECOMMENDATIONS

Based on the findings obtained from the analysis, several recommendations can be proposed to improve instructor performance, course quality, and learner engagement on the EduPro platform.

Recommendation 1: Strengthen Instructor Development Programs

The positive relationship between teaching experience and instructor ratings indicates the importance of professional growth. Regular training programs, workshops, and mentoring initiatives should be introduced to help instructors improve their teaching effectiveness.

Recommendation 2: Promote Best Practices from High-Performing Domains

Expertise areas such as Marketing, Machine Learning, and Programming demonstrated strong performance. The teaching strategies and course structures used in these domains can be studied and replicated across other categories to improve overall educational quality.

Recommendation 3: Enhance Low-Rated Course Categories

Course categories with comparatively lower ratings should be reviewed to identify content gaps, instructional challenges, and learner concerns. Continuous course improvement initiatives can help increase learner satisfaction.

Recommendation 4: Leverage High-Rated Instructors

High-performing instructors should be recognized and encouraged to contribute to curriculum design, faculty mentoring, and content development activities. Their expertise can support platform-wide quality enhancement.

Recommendation 5: Focus on High-Demand Skills

The high enrollment observed in Cybersecurity and Machine Learning courses reflects strong learner demand. Expanding course offerings within these domains can increase platform engagement and attract more learners.

Recommendation 6: Implement Continuous Performance Monitoring

The deployed Streamlit dashboard should be utilized as a continuous monitoring system to track instructor performance, course quality, and enrollment trends, enabling timely interventions and data-driven decision-making.

CONCLUSION

This project successfully developed a comprehensive Instructor Performance and Course Quality Evaluation Framework for the EduPro platform using data analytics and visualization techniques.

The analysis examined instructor demographics, experience, expertise domains, course ratings, enrollment behavior, and course quality indicators. Multiple exploratory data analysis techniques revealed valuable insights regarding the factors influencing educational performance and learner engagement.

The findings demonstrated that instructor experience contributes positively to teaching effectiveness, while expertise domains significantly influence learner satisfaction and enrollment behavior. The analysis also highlighted that course quality is affected by multiple factors beyond instructor ratings, emphasizing the importance of curriculum design and content quality.

To support practical decision-making, an interactive Streamlit dashboard was developed and deployed. The dashboard enables stakeholders to monitor educational performance metrics, explore trends, and make informed decisions through real-time visualizations and filtering capabilities.

Overall, the project provides a scalable and data-driven approach for evaluating educational effectiveness and supporting continuous improvement within online learning platforms.

FUTURE SCOPE

Although the current project provides meaningful insights into instructor performance and course quality, several opportunities exist for future enhancement.

Future Scope 1: Advanced Predictive Analytics

Machine learning models can be developed to predict instructor ratings, learner satisfaction, and future enrollment trends based on historical data patterns.

Future Scope 2: Real-Time Data Integration

Integration with live educational databases and learning management systems can enable real-time performance monitoring and automated reporting.

Future Scope 3: Automated Recommendation Systems

Artificial intelligence techniques can be used to recommend courses, instructors, and learning pathways based on learner preferences and historical behavior.

Future Scope 4: Learner Sentiment Analysis

Student feedback and reviews can be analyzed using Natural Language Processing (NLP) techniques to better understand learner perceptions and identify improvement opportunities.

Future Scope 5: Enhanced Dashboard Functionality

Future versions of the dashboard can include advanced filters, forecasting capabilities, anomaly detection systems, automated alerts, and performance benchmarking features.

Future Scope 6: Institutional Performance Benchmarking

The framework can be expanded to compare performance across multiple institutions, departments, or educational programs, enabling broader educational quality assessment.

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